Methods of proof – 1

The pigeonhole principle: discrete, continuous, probabilistic

Homework

13 сентября 2026 года

Theoretical minimum. In the discrete form the only difficulty is the choice of boxes; in the continuous form the boxes are replaced by measure and the statement becomes Blichfeldt's lemma, from which Minkowski's and Steinhaus' theorems follow; in the probabilistic form the boxes are replaced by an average, and the principle reads "there is an element at least as good as the mean". Three standard box systems are worth remembering: residues modulo N for approximation problems, dyadic scales for problems about orders of magnitude, and the positions in a linear order for monotone subsequences. When the required object is not just present but plentiful, count the bad configurations instead: if their number is smaller than the total, a good one exists.

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1. Prove the simultaneous version of Dirichlet's theorem: for reals $\alpha_1, \dots, \alpha_k$ and every N there is $q \leqslant N^k$ with $\|q\alpha_i\| < \frac{1}{N}$ for all i; deduce Kronecker's theorem on the density of $\{(q\alpha_1, \dots, q\alpha_k)\}$ when $1, \alpha_1, \dots, \alpha_k$ are linearly independent over $\mathbb{Q}$.

2. Prove that among any five points in a unit square two are at distance at most $\frac{\sqrt{2}}{2}$, prove that the optimal constant for n points is of order $n^{-1/2}$, and find the asymptotics of that constant as $n \to \infty$.

3. Prove that there is a power of 2 whose decimal expansion begins with 2026, and describe all sequences of digits that occur as the beginning of a power of 2.

4. Prove the Lebesgue density theorem for a measurable set, and use it to prove Steinhaus' theorem again; explain which of the two proofs generalises to $\mathbb{R}^n$.

5. Let $A \subseteq \{1, \dots, n\}$ have more than $\lceil \frac{n}{2} \rceil$ elements. Prove that A contains two elements one of which divides the other, and that A contains an arithmetic progression of length three when $|A| > \lceil \frac{2n}{3} \rceil$.

6. Prove that any $n + 2$ vectors in $\mathbb{R}^n$ contain two making an angle at most $\frac{\pi}{2}$ with each other, and that 2^n vectors with pairwise obtuse angles cannot exist in $\mathbb{R}^n$ for $n \geqslant 2$.

7. Prove the second-moment refinement: if $\mathrm{E}\, X = m$ and $\operatorname{Var} X \leqslant \sigma^2$, then $X \geqslant m - \lambda\sigma$ with probability at least $1 - \frac{1}{\lambda^2}$; then, using the second and fourth moments of a random cut, show that a graph with m edges contains a bipartite subgraph with at least $\frac{m}{2} + c\sqrt{m}$ edges.

8. Prove the infinite pigeonhole principle: a colouring of an infinite set in finitely many colours has an infinite monochromatic subset; deduce the infinite Ramsey theorem for pairs and explain where compactness would be needed for the finite version.

9. Prove that for every set $A \subseteq \mathbb{Z}$ of positive upper density the difference set $A - A$ contains arbitrarily long arithmetic progressions of the form $\{d, 2d, \dots, kd\}$, or at least prove the case $k = 1$ carefully and state what the general case needs.

10. Prove that in any sequence of $n^2 + 1$ distinct reals one can find $n + 1$ terms forming a monotone subsequence, and prove the sharpness by an explicit example; then give the probabilistic proof of the existence of a long monotone subsequence in a random permutation.